POOJA AWASARE

Civil & Environmental Engineer · Circular Construction & Digital Decision-Support

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Location: Pune, India
Languages: English (professional); learning Dutch

Professional Summary

Civil and environmental engineer focused on the **digital transformation of circular construction**. My work sits where sustainability, digital tools, and construction practice meet: using **BIM, GIS, digital twins, and material-passport data** to make reuse and **reverse-logistics** decisions clearer when availability, quality, location, timing, transport, cost, and CO_2 are uncertain. My Master's thesis on Agile Project Management in the construction industry showed me that transforming construction depends as much on **transparent information flow and stakeholder coordination** as on technical innovation, and led me to a bigger question: what happens to materials at end-of-life, and how do we get reliable information about them to everyone who needs it? That question moved my focus from improving project delivery to **designing data-driven decision-support for circular construction**. With hands-on experience in digital twins, data integration, requirements engineering, and Agile delivery across manufacturers and construction clients, I want to help make circular construction **practical, data-driven, and widely adopted**, which is exactly the aim of the CIRCOLOGIC EngD.

Education

2018–2022 — **HTW Dresden (Hochschule für Technik und Wirtschaft), Germany** — 2.8
Master of Engineering (M.Eng), Environmental Engineering
Thesis: Agile Project Management for Sustainable Construction Innovation

2014–2017 — **Savitribai Phule Pune University, India** — 65.26%
Bachelor of Engineering (B.E), Civil Engineering
Structural design, project planning & environmental studies

2010–2014 — **Maharashtra State Board of Technical Education (MSBTE), India** — 68.11%
Diploma, Civil Engineering

2010 — **Maharashtra State Board (MSBSHSE), India** — 82.55%
10th Standard (Secondary School Certificate)

Professional Experience and Training

Jun’24 – Present — **Career Break** · Belgium & India

- Planned career break spanning relocation from Belgium to India and family. Used the time to deepen expertise in **circular construction and reverse logistics** and to build **GIS / geospatial and machine-learning skills** (Google Earth Engine), preparing for design-science doctoral research.

Aug’23 – Jun’24 — **Civil & Environmental Project Consultant**, **iostring**, Belgium

- Delivered civil projects end to end with a focus on time, cost and value, using project data to guide **material and reuse choices that reduce waste**, treating circularity as part of delivery rather than a side track.
- Defined problem statements, use cases and solution architecture, and linked and analysed project data to support material and reuse decisions.

Nov’22 – Aug’23 — **Project Engineer**, **Hive CPQ**, Ghent, Belgium

- Managed large datasets and **gathered requirements** from manufacturers (Pellenc, Mecatherm, Soproda) and customers, integrating manufacturing, construction and supply-chain data into a data mart.
- Built configurators that **turned complex rules into intuitive digital tools**, translating stakeholder needs into practical solutions for technical and non-technical users.

Jun'22 – Nov'22	**Project Engineer**, **Aug.e**, Brussels, Belgium • Contributed to energy-transition and smart-building projects using a **digital-twin stack** (Microsoft Azure, Navisworks Manage, Neanex, Revit), organising and integrating data from physical assets into digital decision-support. • Worked with IoT, AI and metadata layers linking building and energy data to decisions, close to the data-integration mindset behind **BIM, product passports and logistics parameters**; tracked delivery in Azure DevOps.
Aug'21 – Sep'21	**Trainee (Master's Thesis)**, **HTW Dresden**, Germany • Researched **Agile project planning in an IT-sustainability setting** aligned to UN Global Compact / SDG themes; gathered qualitative and quantitative requirements on environmental impacts (climate, waste, water, energy) and built product backlogs in Jira.
Oct'19	**Trainee**, **Fraunhofer Institute**, Dresden, Germany • Planned a **green business model** for biogas-to-biowax demand and adopted EU sustainability policy, assessing green-business financials and transformation tools.

Technical Skills

Digital & BIM	Revit, Navisworks Manage, Neanex, AutoCAD; digital twins, product passports
Geospatial / GIS	Google Earth Engine, GIS-style mapping, change detection, machine learning
Data & analysis	Data modelling & integration, Tableau, MongoDB, Postman, Lucidchart, Excel
Project & Agile	Jira, Confluence, Asana, Miro, Azure DevOps, Primavera P6; Scrum, Kanban (PSK I)
Domain	Circular construction, reverse logistics, LCA / CO_2 awareness, EU sustainability policy, UN SDGs
Research methods	Design-science research, qualitative case studies, requirements engineering

Certifications

- Professional Scrum with Kanban (PSK I) · Scrum.org
- Google Earth Engine for Machine Learning & Change Detection
- Construction Management: Planning and Scheduling
- Primavera P6
- Excel Essential Training (Office 365 / Microsoft 365)

Publications

"Design and Analysis of Building by AKSES and Manual Method."

Areas of Interest

- Circular construction: reuse, refurbishment and end-of-life material recovery
- Reverse logistics and construction supply chains
- Digital decision-support: BIM, GIS, digital twins and material passports
- Design-science research for sustainability and CO_2 reduction

References

Frederik Taleman
CEO, Hive-CPQ

Prof. Dr. rer. nat. Martin Oczipka
HTW Dresden, Germany

Micheal Dhont
Consultant, McKinsey & Company; former Product Head, Aug.e

Khushal Kanade
Assistant Professor, MIT, Pune, India

Full academic transcript (courses and grades) available on request.